

TRANSFORMER

Numerical Problems with Complete Solutions

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Subject: Electrical Science — Unit 4

Q1. EMF Equation of Transformer

The primary winding of a 50 Hz transformer is supplied from a 440 V, 50 Hz source and has 200 turns. Find:

- (i) Peak value of flux
- (ii) Voltage induced in the secondary winding if it has 50 turns.

◆ SOLUTION

Given:

$$f = 50 \text{ Hz}, E_1 = 440 \text{ V}, N_1 = 200, N_2 = 50$$

Part (i): Peak value of flux (ϕ_m)

$$\text{Using EMF equation: } E_1 = 4.44 \times f \times \phi_m \times N_1$$

$$\phi_m = E_1 / (4.44 \times f \times N_1)$$

$$\phi_m = 440 / (4.44 \times 50 \times 200)$$

$$\phi_m = 440 / 44400$$

$$\therefore \phi_m = 0.0099 \text{ Wb} \approx 9.9 \text{ mWb}$$

Part (ii): Voltage induced in secondary (E_2)

$$E_2 = 4.44 \times f \times \phi_m \times N_2$$

$$E_2 = 4.44 \times 50 \times 0.0099 \times 50$$

$$\therefore E_2 = 110 \text{ V}$$

Q2. No-Load Operation — Example 3.3

The no-load current of a transformer is 10 A at a power factor of 0.25 lagging, when connected to 400 V, 50 Hz supply. Primary winding turns = 500. Calculate:

- (a) Magnetising component of no-load current
- (b) Iron loss
- (c) Maximum value of flux in the core

◆ SOLUTION

Given:

$$I_o = 10 \text{ A}, \cos \phi_o = 0.25, V_1 = 400 \text{ V}, f = 50 \text{ Hz}, N_1 = 500$$

(a) Magnetising component (I_m):

$$\phi_o = \cos^{-1}(0.25) = 75.522^\circ$$

$$I_m = I_o \times \sin \phi_o = 10 \times \sin(75.522^\circ)$$

$$\therefore I_m = 10 \times 0.96824 = 9.6824 \text{ A}$$

(b) Iron Loss (P_i):

$$P_i = W_o = V_1 \times I_o \times \cos \phi_o$$

$$P_i = 400 \times 10 \times 0.25$$

$$\therefore P_i = 1000 \text{ W} = 1 \text{ kW}$$

(c) Maximum flux (ϕ_m):

$$\text{On no load: } E_1 = V_1 = 400 \text{ V}$$

$$E_1 = 4.44 \times f \times \phi_m \times N_1$$

$$400 = 4.44 \times 50 \times \phi_m \times 500$$

$$\phi_m = 400 / (4.44 \times 50 \times 500) = 400 / 111000$$

$$\therefore \phi_m = 3.6036 \text{ mWb}$$

Q3. Transformer on Load — Primary Current Calculation

A 400/200 V transformer takes 1 A at a power factor of 0.4 on no load. If the secondary supplies a load current of 50 A at 0.8 lagging power factor, calculate the primary current.

◆ SOLUTION

Given:

$$I_o = 1 \text{ A}, \cos \phi_o = 0.4, I_2 = 50 \text{ A}, \cos \phi_2 = 0.8 \text{ (lagging)}$$

$$V_1 = 400 \text{ V}, V_2 = 200 \text{ V}$$

Step 1: Find transformation ratio K

$$K = E_2/E_1 = 200/400 = 0.5$$

Step 2: Load component of primary current

$$I'_2 = K \times I_2 = 0.5 \times 50 = 25 \text{ A}$$

$$\phi_2 = \cos^{-1}(0.8) = 36.86^\circ \text{ (} I_2 \text{ lags by } 36.86^\circ, \text{ so } I'_2 \text{ is antiphase)}$$

Step 3: Resolve no-load current I_o

$$\phi_o = \cos^{-1}(0.4) = 66.42^\circ$$

$$I_o \text{ (x-component)} = I_o \sin \phi_o = 1 \times \sin(66.42^\circ) = 0.9165 \text{ A}$$

$$I_o \text{ (y-component)} = I_o \cos \phi_o = 1 \times 0.4 = 0.4 \text{ A}$$

$$\therefore I_o = 0.9165 + j 0.4 \text{ A}$$

Step 4: Resolve I'_2

$$I'_2 \text{ (x-component)} = 25 \times \sin(36.86^\circ) = 25 \times 0.6 = 15 \text{ A}$$

$$I'_2 \text{ (y-component)} = 25 \times \cos(36.86^\circ) = 25 \times 0.8 = 20 \text{ A}$$

$$\therefore I'_2 = 15 + j 20 \text{ A}$$

Step 5: Primary current $I_1 = I'_2 + I_o$

$$I_1 = (15 + 0.9165) + j(20 + 0.4) = 15.9165 + j 20.4 \text{ A}$$

$$|I_1| = \sqrt{(15.9165^2 + 20.4^2)} = \sqrt{(253.33 + 416.16)} = \sqrt{669.49}$$

$$\therefore I_1 = 25.874 \text{ A}$$

$$\text{Power factor: } \cos \phi_1 = \cos(37.96^\circ) = 0.788 \text{ lagging}$$

Q4. Equivalent Resistance — Example 3.5

A 6600/400 V single phase transformer has primary resistance $R_1 = 2.5 \Omega$ and secondary resistance $R_2 = 0.01 \Omega$. Calculate total equivalent resistance referred to primary and secondary.

◆ SOLUTION

Given:

$$V_1 = 6600 \text{ V}, V_2 = 400 \text{ V}, R_1 = 2.5 \Omega, R_2 = 0.01 \Omega$$

Transformation ratio:

$$K = V_2/V_1 = 400/6600 = 0.0606$$

Equivalent resistance referred to PRIMARY (R_{1e}):

$$R'_2 = R_2/K^2 = 0.01/(0.0606)^2 = 0.01/0.003672 = 2.7225 \Omega$$

$$R_{1e} = R_1 + R'_2 = 2.5 + 2.7225$$

$$\therefore R_{1e} = 5.2225 \Omega$$

Equivalent resistance referred to SECONDARY (R_{2e}):

$$R'_1 = K^2 \times R_1 = (0.0606)^2 \times 2.5 = 0.003672 \times 2.5 = 0.00918 \Omega$$

$$R_{2e} = R_2 + R'_1 = 0.01 + 0.00918$$

$$\therefore R_{2e} = 0.01918 \Omega$$

Q5. Equivalent Impedance — Example 3.6

A 220/110 V, 50 Hz, 1.5 kVA transformer has primary and secondary winding resistances of 1 Ω and 2 Ω while reactances of 3 Ω and 5 Ω respectively. Find the total resistance, equivalent reactance and equivalent impedance referred to primary and secondary.

◆ SOLUTION

Given:

$$R_1 = 1 \, \Omega, R_2 = 2 \, \Omega, X_1 = 3 \, \Omega, X_2 = 5 \, \Omega$$

$$K = V_2/V_1 = 110/220 = 0.5$$

Referred to PRIMARY:

$$R_{1e} = R_1 + R_2/K^2 = 1 + 2/(0.5)^2 = 1 + 2/0.25 = 1 + 8 = 9 \, \Omega$$

$$X_{1e} = X_1 + X_2/K^2 = 3 + 5/(0.5)^2 = 3 + 20 = 23 \, \Omega$$

$$Z_{1e} = \sqrt{(R_{1e})^2 + (X_{1e})^2} = \sqrt{(9)^2 + (23)^2} = \sqrt{81 + 529} = \sqrt{610}$$

$$\therefore Z_{1e} = 24.698 \, \Omega, R_{1e} = 9 \, \Omega, X_{1e} = 23 \, \Omega$$

Referred to SECONDARY:

$$R_{2e} = K^2 \times R_{1e} = (0.5)^2 \times 9 = 0.25 \times 9 = 2.25 \, \Omega$$

$$X_{2e} = K^2 \times X_{1e} = (0.5)^2 \times 23 = 0.25 \times 23 = 5.75 \, \Omega$$

$$Z_{2e} = K^2 \times Z_{1e} = (0.5)^2 \times 24.698 = 6.1745 \, \Omega$$

$$\therefore Z_{2e} = 6.1745 \, \Omega, R_{2e} = 2.25 \, \Omega, X_{2e} = 5.75 \, \Omega$$

Q6. Voltage Regulation

A 250/125 V, 5 kVA single phase transformer has primary resistance 0.2 Ω , reactance 0.75 Ω ; secondary resistance 0.05 Ω , reactance 0.2 Ω .

- Determine its regulation while supplying full load on 0.8 leading p.f.
- The secondary terminal voltage on full load and 0.8 leading p.f.

◆ SOLUTION

Given:

$$R_1 = 0.2 \, \Omega, X_1 = 0.75 \, \Omega, R_2 = 0.05 \, \Omega, X_2 = 0.2 \, \Omega, \cos \phi = 0.8 \text{ leading}$$

$$K = 125/250 = 0.5; \text{ Full load } I_2 = \text{kVA}/V_2 = 5000/125 = 40 \, \text{A}$$

Equivalent parameters referred to secondary:

$$R_{2e} = R_2 + K^2 R_1 = 0.05 + (0.5)^2 \times 0.2 = 0.05 + 0.05 = 0.1 \, \Omega$$

$$X_{2e} = X_2 + K^2 X_1 = 0.2 + (0.5)^2 \times 0.75 = 0.2 + 0.1875 = 0.3875 \, \Omega$$

(i) % Voltage Regulation (leading load $\rightarrow$ minus sign):

$$\sin \phi = 0.6 \text{ (since } \cos \phi = 0.8)$$

$$\% R = [(I_2 R_{2e} \cos \phi - I_2 X_{2e} \sin \phi) / E_2] \times 100$$

$$\% R = [(40 \times 0.1 \times 0.8 - 40 \times 0.3875 \times 0.6) / 125] \times 100$$

$$\% R = [(3.2 - 9.3) / 125] \times 100 = (-6.1/125) \times 100$$

$$\therefore \% \text{ Regulation} = -4.88\% \text{ (Negative } \rightarrow \text{ leading p.f.)}$$

(ii) Secondary terminal voltage on full load:

$$\text{Voltage drop} = I_2 [R_{2e} \cos \phi - X_{2e} \sin \phi] = 40[0.1 \times 0.8 - 0.3875 \times 0.6] = 40[0.08 - 0.2325] = -6.1 \, \text{V}$$

$$E_2 - V_2 = -6.1 \rightarrow 125 - V_2 = -6.1$$

$$\therefore V_2 = 131.1 \, \text{V}$$

Q7. Efficiency of Transformer

A 4 kVA, 200/400 V, 50 Hz single phase transformer has equivalent resistance referred to primary $R_{1e} = 0.15 \Omega$. Iron losses $P_i = 60$ W. Calculate:

- (i) Total copper losses on full load
- (ii) Efficiency while supplying full load at 0.9 p.f. lagging
- (iii) Efficiency while supplying half load at 0.8 p.f. leading

◆ SOLUTION

Given:

$$\text{kVA} = 4, V_1 = 200 \text{ V}, V_2 = 400 \text{ V}, R_{1e} = 0.15 \Omega, P_i = 60 \text{ W}$$

$$K = 400/200 = 2; R_{2e} = K^2 \times R_{1e} = 4 \times 0.15 = 0.6 \Omega$$

$$\text{Full load } I_2 = \text{kVA}/V_2 = 4000/400 = 10 \text{ A}$$

(i) Copper losses at full load:

$$P_{cu}(\text{FL}) = I_2^2 \times R_{2e} = (10)^2 \times 0.6 = 100 \times 0.6$$

$$\therefore P_{cu}(\text{FL}) = 60 \text{ W}$$

(ii) Efficiency at full load, $\cos \phi = 0.9$ lagging:

$$\text{Output} = VA \times \cos \phi = 4000 \times 0.9 = 3600 \text{ W}$$

$$\text{Total losses} = P_i + P_{cu} = 60 + 60 = 120 \text{ W}$$

$$\eta = \text{Output} / (\text{Output} + \text{Losses}) \times 100$$

$$\eta = 3600 / (3600 + 120) \times 100 = 3600/3720 \times 100$$

$$\therefore \eta = 96.77\%$$

(iii) Efficiency at half load, $\cos \phi = 0.8$ leading:

$$n = 0.5 \text{ (half load)}$$

$$\text{New } P_{cu} = n^2 \times P_{cu}(\text{FL}) = (0.5)^2 \times 60 = 0.25 \times 60 = 15 \text{ W}$$

$$\text{Output} = n \times VA \times \cos \phi = 0.5 \times 4000 \times 0.8 = 1600 \text{ W}$$

$$\eta = 1600 / (1600 + 60 + 15) \times 100 = 1600/1675 \times 100$$

$$\therefore \eta = 95.52\%$$

Q8. Maximum Efficiency — kVA at Maximum Efficiency

A 100 kVA transformer has iron losses of 1.2 kW and full load copper losses of 1.5 kW. Find:

- (i) kVA for maximum efficiency
- (ii) Maximum efficiency at unity p.f.

◆ SOLUTION

Given:

$$\text{kVA} = 100, P_i = 1.2 \text{ kW}, P_{cu}(\text{FL}) = 1.5 \text{ kW}, \cos \phi = 1$$

(i) kVA at maximum efficiency:

Condition for η_{max} : Copper losses = Iron losses

$$kVA \text{ at } \eta_{max} = kVA(\text{rated}) \times \sqrt{P_i / P_{cu}(\text{FL})}$$

$$kVA \text{ at } \eta_{max} = 100 \times \sqrt{(1.2/1.5)} = 100 \times \sqrt{0.8} = 100 \times 0.8944$$

$$\therefore kVA \text{ at } \eta_{max} = 89.44 \text{ kVA}$$

(ii) Maximum efficiency:

$$\text{At } \eta_{max}, P_{cu} = P_i = 1.2 \text{ kW}$$

$$\text{Output at } \eta_{max} = 89.44 \times 10^3 \times 1 = 89440 \text{ W}$$

$$\text{Total losses} = 2 \times P_i = 2 \times 1200 = 2400 \text{ W}$$

$$\eta_{max} = 89440 / (89440 + 2400) \times 100 = 89440/91840 \times 100$$

$$\therefore \eta_{max} = 97.39\%$$

Q9. Efficiency at Full and Half Load — Example 8.21

In a 25 kVA, 2000/200 V transformer the iron and full load copper losses are 350 W and 400 W respectively. Find the efficiency at unity p.f. at (a) full load (b) half load. Also determine the load for maximum efficiency.

◆ SOLUTION

Given:

$$kVA = 25, P_i = 350 \text{ W}, P_{cu}(\text{FL}) = 400 \text{ W}, \cos \phi = 1$$

(a) Efficiency at full load and unity p.f.:

$$\text{Output} = 25 \times 10^3 \times 1 = 25000 \text{ W}$$

$$\text{Total losses} = P_i + P_{cu}(\text{FL}) = 350 + 400 = 750 \text{ W}$$

$$\eta = 25000 / (25000 + 750) \times 100 = 25000/25750 \times 100$$

$$\therefore \eta (\text{FL}) = 97.09\% \approx 97\%$$

(b) Efficiency at half load and unity p.f.:

$$n = 0.5, \text{Output} = 0.5 \times 25000 = 12500 \text{ W}$$

$$P_{cu}(\text{HL}) = n^2 \times P_{cu}(\text{FL}) = (0.5)^2 \times 400 = 0.25 \times 400 = 100 \text{ W}$$

$$\text{Total losses} = 350 + 100 = 450 \text{ W}$$

$$\eta = 12500 / (12500 + 450) \times 100 = 12500/12950 \times 100$$

$$\therefore \eta (\text{HL}) = 96.53\% \approx 96.5\%$$

Load for maximum efficiency:

$$\text{At } \eta_{max}, P_{cu} = P_i \rightarrow x^2 \times 400 = 350$$

$$x^2 = 350/400 = 0.875 \rightarrow x = \sqrt{0.875} = 0.935$$

$$\text{Load at } \eta_{max} = x \times kVA = 0.935 \times 25 = 23.375 \text{ kVA}$$

$$\therefore \text{Load for maximum efficiency} = 23.375 \text{ kVA}$$

Q10. All Day Efficiency — Example 8.37

A transformer has its maximum efficiency of 0.975 at 20 kVA at unity p.f. During the day it is loaded as follows:

10 hr: 3 kW at 0.6 p.f. | 8 hr: 10 kW at 0.8 p.f. | 6 hr: 20 kW at 0.9 p.f.

Find the all day efficiency.

◆ SOLUTION

Step 1: Total energy output (kWh)

$$\text{Output} = (10 \times 3) + (8 \times 10) + (6 \times 20) = 30 + 80 + 120 = 230 \text{ kWh}$$

Step 2: Find P_i and P_{cu} at $\eta_{\max}$

$$\text{At } \eta_{\max} = 0.975: \text{Total losses} = (1 - 0.975) / 0.975 \times \text{output_power} \approx 0.025 \times 20000 = 500 \text{ W}$$

$$\text{Core losses} = \text{Copper losses} = 500 / 2 = 250 \text{ W each}$$

$$(\text{Rating} = 20 \text{ kVA at } \eta_{\max}, \text{ so } P_{cu}(\max) = 250 \text{ W at 20 kVA})$$

Step 3: Total core loss energy over 24 hr (constant)

$$\text{Core loss energy} = 250 \times 24 / 1000 = 6 \text{ kWh}$$

Step 4: Copper loss energy for each period

$$\text{For 10 hr period: kVA load} = 3 / 0.6 = 5 \text{ kVA}$$

$$P_{cu} = (5/20)^2 \times 250 = 0.0625 \times 250 = 15.625 \text{ W}$$

$$\text{Cu loss energy} = 10 \times 15.625 / 1000 = 0.156 \text{ kWh}$$

$$\text{For 8 hr period: kVA load} = 10 / 0.8 = 12.5 \text{ kVA}$$

$$P_{cu} = (12.5/20)^2 \times 250 = 0.390625 \times 250 = 97.66 \text{ W}$$

$$\text{Cu loss energy} = 8 \times 97.66 / 1000 = 0.781 \text{ kWh}$$

$$\text{For 6 hr period: kVA load} = 20 / 0.9 = 22.22 \text{ kVA}$$

$$P_{cu} = (22.22/20)^2 \times 250 = 1.2345 \times 250 = 308.6 \text{ W}$$

$$\text{Cu loss energy} = 6 \times 308.6 / 1000 = 1.852 \text{ kWh}$$

$$\text{Total copper loss energy} = 0.156 + 0.781 + 1.852 = 2.789 \text{ kWh}$$

$$\text{Total energy loss} = 6 + 2.789 = 8.789 \text{ kWh}$$

All day efficiency:

$$\eta_{\text{all-day}} = \text{Output} / (\text{Output} + \text{Losses}) = 230 / (230 + 8.789) = 230 / 238.789$$

$$\therefore \eta_{\text{all-day}} = \mathbf{96.32\%}$$

Q11. All Day Efficiency — Example 8.38

A lighting transformer rated at 10 kVA has full load losses of 0.3 kW, made up equally from iron losses and copper losses. Duty cycle: full load for 3 hours, half full load for 4 hours, no load for remainder of 24 hours. Load operates at unity power factor. Calculate the all day efficiency.

❖ SOLUTION

Given:

kVA = 10, Total FL losses = 0.3 kW, $\cos \phi = 1$

$$P_i = P_{cu}(\text{FL}) = 0.3/2 = 0.15 \text{ kW} = 150 \text{ W each}$$

Step 1: Total energy output

Full load (3 hr): Output = $10 \times 1 \times 3 = 30 \text{ kWh}$

Half load (4 hr): Output = $(1/2) \times 10 \times 1 \times 4 = 20 \text{ kWh}$

Total output = $30 + 20 = 50 \text{ kWh}$

Step 2: Core (iron) loss energy over full 24 hr

Iron loss is constant at all times (even no-load)

Core loss energy = $0.15 \times 24 = 3.6 \text{ kWh}$

Step 3: Copper loss energy

Full load (3 hr): $P_{cu} = 0.15 \text{ kW} \rightarrow \text{Energy} = 0.15 \times 3 = 0.45 \text{ kWh}$

Half load (4 hr): $P_{cu} = (1/2)^2 \times 0.15 = 0.0375 \text{ kW} \rightarrow \text{Energy} = 0.0375 \times 4 = 0.15 \text{ kWh}$

No load (17 hr): $P_{cu} = 0 \rightarrow \text{Energy} = 0 \text{ kWh}$

Total copper loss energy = $0.45 + 0.15 = 0.60 \text{ kWh}$

Step 4: Total losses and efficiency

Total energy loss = $3.6 + 0.6 = 4.2 \text{ kWh}$

$$\eta_{\text{all-day}} = \text{Output} / (\text{Output} + \text{Losses}) = 50 / (50 + 4.2) = 50/54.2$$

$$\therefore \eta_{\text{all-day}} = \mathbf{92.25\%}$$

SUMMARY OF ANSWERS

Q No.	Topic	Key Answer(s)
1	EMF Equation	$\phi_m = 9.9 \text{ mWb}$, $E_2 = 110 \text{ V}$
2	No-Load Operation	$I_m = 9.68 \text{ A}$, $P_i = 1 \text{ kW}$, $\phi_m = 3.60 \text{ mWb}$
3	Transformer on Load	$I_1 = 25.87 \text{ A}$, p.f. = 0.788 lagging
4	Equivalent Resistance	$R_{1e} = 5.22 \Omega$, $R_{2e} = 0.0192 \Omega$
5	Equivalent Impedance	$Z_{1e} = 24.70 \Omega$, $Z_{2e} = 6.17 \Omega$
6	Voltage Regulation	%R = -4.88%, $V_2 = 131.1 \text{ V}$
7	Efficiency	$\eta(\text{FL}) = 96.77\%$, $\eta(\text{HL}) = 95.52\%$
8	Maximum Efficiency	89.44 kVA, $\eta_{\text{max}} = 97.39\%$
9	Full & Half Load η	$\eta(\text{FL}) = 97\%$, $\eta(\text{HL}) = 96.5\%$, Load = 23.375 kVA
10	All Day Efficiency	$\eta_{\text{all-day}} = 96.32\%$

11	All Day Efficiency	$\eta_{\text{all-day}} = 92.25\%$
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